

Exercise: Tuples and Sets

Please, submit your source code solutions for the described problems to the [Judge System](#).

Ask your questions here: <https://www.slido.com> by entering the course code **#python-advanced**

1. Unique Usernames

Write a program that reads a sequence of **N usernames** from the console and maintains a collection of only the **unique** ones.

On the **first** line, you will receive an integer **N**.

On the following **N** lines, you will receive the **usernames**. Print the collection on the console (the order does **not** matter):

Examples

Input	Output
6 George George George Peter George NiceGuy1234	George Peter NiceGuy1234
10 Peter Maria Peter George Steve Maria Alex Peter Steve George	Peter Maria George Steve Alex

2. Sets of Elements

Write a program that prints a **set of elements**. On the **first line**, you will receive **two numbers** - **n** and **m**, separated by a single space - representing the lengths of two separate sets. On the next **n + m** lines, you will receive **n** numbers, which are the numbers in the **first** set, and **m** numbers, which are in the **second** set. Find all the **unique elements** that appear in **both** and **print** them on **separate lines** (the order **does not** matter).

For example:

Set with length n = 4: {1, 3, 5, 7}

Set with length $m = 3$: {3, 4, 5}

Set that contains all the **elements** that repeat in **both sets** -> {3, 5}

Examples

Input	Output
4 3 1 3 5 7 3 4 5	3 5
2 2 1 3 1 5	1

3. Periodic Table

Write a program that keeps all the **unique** chemical **elements**. On the first line, you will be given a number **n** - the **count** of input **lines** that you will receive. On the following **n** lines, you will be receiving **chemical compounds** separated by a **single space**. Your task is to print all the **unique ones** on separate lines (the **order does not matter**):

Examples

Input	Output
4 Ce O Mo O Ce Ee Mo	Ce Ee Mo O
3 Ge Ch O Ne Nb Mo Tc O Ne	Ch Ge Mo Nb Ne O Tc

4. Count Symbols

Write a program that reads a **text** from the console and **counts** the **occurrences** of **each** character in it. Print the results in **alphabetical** (lexicographical) order.

Examples

Input	Output	Input	Output
SoftUni rocks	: 1 time/s S: 1 time/s U: 1 time/s c: 1 time/s f: 1 time/s i: 1 time/s k: 1 time/s n: 1 time/s o: 2 time/s r: 1 time/s s: 1 time/s t: 1 time/s	Why do you like Python?	: 4 time/s ?: 1 time/s P: 1 time/s W: 1 time/s d: 1 time/s e: 1 time/s h: 2 time/s i: 1 time/s k: 1 time/s l: 1 time/s n: 1 time/s o: 3 time/s t: 1 time/s u: 1 time/s y: 3 time/s

5. Longest Intersection

Write a program that finds the **longest intersection**. You will be given a number **N**. On each of the next **N** lines you will be given **two ranges** in the format: "{**first_start**},{**first_end**}-{**second_start**},{**second_end**}". You should find the **intersection** of these two ranges. The **start** and **end numbers** in the ranges are **inclusive**.

Finally, you should **find the longest intersection of all N** intersections, print the **numbers** that are included and its length in the format: "Longest intersection is [{**longest_intersection_numbers**}] with length {**length_longest_intersection**}"

Note: in each range, there will always be an intersection. If there are two equal intersections, print the first one.

Examples

Input	Output	Comment
3 0,3-1,2 2,10-3,5 6,15-3,10	Longest intersection is [6, 7, 8, 9, 10] with length 5	The intersection of [0-3] and [1-2] is [1-2] (length 2) The intersection of [2-10] and [3-5] is [3-5] (length 3) The intersection of [6-15] and [3-10] is [6-10] (length 5) - which is the longest
5 0,10-2,5	Longest intersection is [2, 3, 4, 5, 6, 7, 8, 9, 10] with length 9	

3,8-1,7 1,8-2,4 4,7-2,5 1,10-2,11		
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6. Battle of Names [Solve with AI]

You will receive a **number N**. On the following **N** lines, you will be receiving **names**. You should **sum the ASCII values** of each letter in the name and **integer divide it by the number of the current row (starting from 1)**. Save the result to a set of **either odd or even** numbers, depending on whether the resulting number is **odd or even**. After that, **sum the values of each set**.

- If the **sums of the two sets are equal**, print the **union of the values**, separated by ", ".
- If the **sum of the odd numbers is bigger than the sum of the even numbers**, print the **different values**, separated by ", ".
- If the **sum of the even numbers is bigger than the sum of the odd numbers**, print the **symmetric-different values**, separated by ", ".

NOTE: On every operation, the starting set should be the odd set

Examples

Input	Output	Comment
4 Pesho Stefan Stamat Gosho	304, 128, 206, 511	First name: Pesho. The sum of the ASCII values is $80 + 101 + 115 + 104 + 111 = 511$. Integer divides the sum to the current row (1): $511 / 1 = 511$. Second name: Stefan. The sum of the ASCII values is $83 + 116 + 101 + 102 + 97 + 110 = 609$. Integer divides the sum to the current row (2): $609 / 2 = 304$. Third name: Stamat. The sum of the ASCII values is $83 + 116 + 97 + 109 + 97 + 116 = 618$. Integer divides the sum to the current row (3): $618 / 3 = 206$. Fourth name: Gosho. The sum of the ASCII values is $71 + 111 + 115 + 104 + 111 = 512$. Integer divides the sum to the current row (4): $512 / 4 = 128$. The odd set: 511 The even set: 304, 206, 128 The sum of the even numbers is larger, so we print the symmetric-different values.
6 Preslav Gosho Ivan Stamat Pesho Stefan	733, 101	